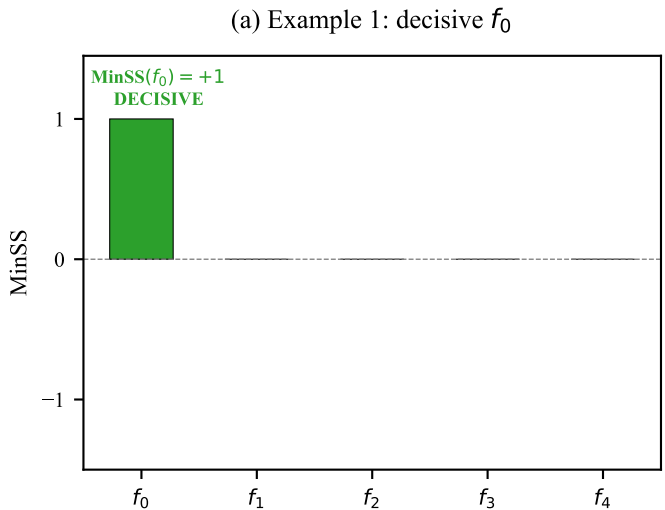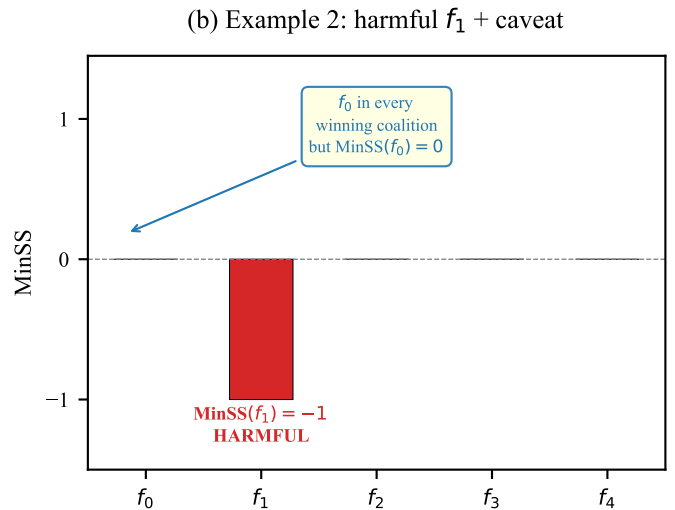Figure 1: MinPPI Trichotomy — Lemma 1 & Theorem 1